Integrate with EasyPark

Import and manage external permit data API

/api/permit

EasyPark AB
Positionen 103
115 74 Stockholm

V. 1.1
October 01, 2023

Definitions

Permit HUB	A cloud-based application that stores and gives secure access to permit applications with their corresponding configurations to support usage in a variety of parking applications
Smart HUB	A cloud-based application that collects parkings in one common way. It allows the establishment of a fully digitalized system, where every single component of the parking operation is able to communicate by sending or receiving data to/from the HUB-server in real-time. All information about parking transactions, no matter if it goes about short-term parking or parking permits, are securely stored in this hub and are made available only to authorized users (e.g. Enforcement, Analytics tools etc.)
Permit application	An exemption to the standard parking policy which defines different rules, for example who has a right to acquire; what is the price policy; which areas in the city are included, among others. Each instance applies to a specific user and set of parking areas.

Introduction

This is the REST/JSON API for adding data related to permit applications to EasyPark Permit HUB. This API is used when an external permit system issues a new permit application for users who will use the EasyPark app to make use of this permit, via subtickets or discounted rates.

For the current version of this API and a testing environment, go to this link:
https://external-gw-staging.easyparksystem.net/api/swagger-ui.html

Purpose

The purpose of this document is to describe how an external party (e.g. Parking permit system) could post information to the EP platform about permit application being issued to an user.

Area settings

For a parking area to work with the Permit HUB it has to be configured in two possible ways:

- Allow permit HUB discounts: this area will allow users that have a permit application to park at a discounted price. *Internal configuration name: include permit hub discounts.*
- Require permit HUB application: this area will not allow parking for users that do not have a permit application, so only users that have a permit application will be able to start parkings

in it. Since this is a subset of cases, any area with this configuration will also allow permit HUB discounts. *Internal configuration name: require permit hub functionality.*

Permit Application options

By using **Permit HUB** users can create 3 types of permit applications, by combining various arguments. Types are:

- ***Non activated permit:*** Permit holder can park in the designated area(s) without manually starting a parking. Until the application expires (as defined in the `validTo` field).

  Relevant API parameters for this application:
  - `visibleForEnforcement = true`
  - `discountRate = null`
  - `remainingParkingMinutes = null`

- ***Activated unlimited permit:*** Permit holder can park in the designated area(s) for a discounted rate (up to 100% discount which would make the parking free), has to manually start a parking. Until the application expires (as defined in the `validTo` field).

  Relevant API parameters for this application:
  - `visibleForEnforcement = false`
  - `discountRate >= 0`
  - `remainingParkingMinutes = null`

- ***Activated limited permit*:** Permit holder can park in the designated area(s) for a discounted rate (up to 100% discount which would make the parking free) for a certain amount of time (as defined in `remainingParkingMinutes`), has to manually start a parking. Until the application expires (as defined in the `validTo` field).

  Relevant API parameters for this application:
  - `visibleForEnforcement = false`
  - `discountRate >= 0`
  - `remainingParkingMinutes >= 0`

Environments & Security

There are 2 environments: staging for testing, and production. To access the APIs, the user has to perform a login by making the following http request and replacing the refresh token with a token

that will be provided by EasyPark with relevant access. Refresh tokens will be different in staging and production environments.

Developer portal link:

https://external-gw-staging.easyparksystem.net/api/swagger-ui.html#/authentication-resource/getJwtUsingPOST

Endpoints:

Staging	https://external-gw-staging.easyparksystem.net/api/get-jwt
Production	https://external-gw.easyparksystem.net/api/get-jwt

Method: `POST`

Header:

```
Content-Type: application/json
```

Sample Request:

```
{ "refreshToken": "dummy_token_that_will_be_provided" }
```

The refresh token will be provided by EasyPark.

Sample Response:

```
{ "validUntilDate": 2021-05-12T07:40:38.656+0000Z, "idToken": "idtoken"
}
```

Given `idToken` will be valid until the date of `validUntilDate`, after this date, the user has to perform a login action again with the original `refreshToken`. Keep in mind to update the token a few minutes before it expires, to avoid the situation where the token expires during the request.

Common misunderstanding: `refreshToken` *is not an* `idToken`*.*

API stack

For the current version of this API stack and a testing environment, go to this link: https://external-gw-staging.easyparksystem.net/api/swagger-ui.html

Create a permit application

Developer portal link:

https://external-gw-staging.easyparksystem.net/api/swagger-ui.html#/permit-resource/createPermitUsingPOST

Endpoints:

Staging	https://external-gw-staging.easyparksystem.net/api/permit/create
Production	https://external-gw.easyparksystem.net/api/permit/create

Method: `POST`

Header:

```
Content-Type: application/json
X-Authorization: Bearer idToken
```

Request JSON object

Parameter name	Required?	Type	Description
permitName	Yes	string	Permit name to be shown in user interaction.
validFrom	Yes	string	Datetime that indicates when the user is able to start utilizing the permit. Format is yyyy-MM-ddTHH:mm:ss.SSSZ (ISO 8601)
validTo	Yes	string	Datetime that indicates when the permit expires. Format is yyyy-MM-ddTHH:mm:ss.SSSZ (ISO 8601)
carLicenseNumber	No*	string	License number without spaces or special characters, only one license plate per permit application. Either carLicenseNumber or phoneNumber is required to match the permit application to a vehicle or user.
phoneNumber	No*	string	Format: +COUNTRYCODE and no leading zeros. Either carLicenseNumber or phoneNumber is required to match the permit application to a vehicle or user.

permitIdentifier	Yes	string	Two possible values are “USE_CARLICENSENUMBER” and “USE_PHONENUMBER” depending on how the application is to be found. If both possibilities are required, there can be 2 identical applications, one with car license and the other with phone number.
parkingareaNumbers	Yes	Array	Array of parkingareaNumber objects described below.
visibleForEnforcement	Yes	boolean	Value should be true for non-activated permits and false for activated permits.
discountRate	Yes	decimal	From 0 to 100, percentage value of discounted parking. Has to be set to null for non activated permits.
maxDiscount	No	decimal	Monetary value of the maximum discount that can be applied with the discountRate. Only relevant for activated permits.
remainingParkingMinutes	No*	integer	Has to be set to a value >= 0 for activated limited permit, otherwise should be null.
priceExclVat	No	decimal	Price without VAT
priceNotSubjectToVat	No	decimal	Price not subject to VAT
priceTotal	No	decimal	Price total including VAT
priceVat	No	decimal	VAT value
permitNote	No	string	Notes associated with permit in case they are necessary.

parkingareaNumber object

Parameter name	Required?	Type	Description
areaCountryCode	Yes	string	2-letter country code (ISO 3166-1 alpha-2)
areaNo	Yes	integer	Area code

Response JSON object

This will create a separate permit application for every area defined in the parameter `parkingareaNumbers`, and so the response will be an array of these permit applications. Each

object will include all fields back as verification, and the important addition of the field `uniqueId`, which will be used for reference in the Delete, Get, and CarSwitch methods; and the field `parkingareaNo` which is how the area is stored in the HUB database.

Parameter name	Required?	Type	Description
uniqueId	Yes	string	Unique identifier to be used in further operations of the Permit HUB
permitName	Yes	string	Permit name to be shown in user interaction.
validFrom	Yes	string	Datetime that indicates when the user is able to start utilizing the permit. Format is yyyy-MM-ddTHH:mm:ss.SSSZ (ISO 8601)
validTo	Yes	string	Datetime that indicates when the permit expires. Format is yyyy-MM-ddTHH:mm:ss.SSSZ (ISO 8601)
carLicenseNumber	No*	string	License number without spaces or special characters, only one license plate per permit application. Either carLicenseNumber or phoneNumber is required to match the permit application to a vehicle or user.
phoneNumber	No*	string	Format: +COUNTRYCODE and no leading zeros. Either carLicenseNumber or phoneNumber is required to match the permit application to a vehicle or user.
permitIdentifier	Yes	string	Two possible values are “USE_CARLICENSENUMBER” and “USE_PHONENUMBER” depending on how the application is to be found.
parkingareaNo	Yes	integer	HUB ID for parking area applicable for this permit application.
visibleForEnforcement	Yes	boolean	Value will be true for non-activated permits and false for activated permits.
discountRate	Yes	decimal	From 0 to 100, percentage value of discounted parking. Will be null for non activated permits.
maxDiscount	No	decimal	Monetary value of the maximum discount that can be applied with the discountRate. Only relevant for activated permits.

remainingParkingMinutes	No*	integer	Will be set to a value >= 0 for activated limited permit, otherwise will be null.
priceExclVat	No	decimal	Price without VAT
priceNotSubjectToVat	No	decimal	Price not subject to VAT
priceTotal	No	decimal	Price total including VAT
priceVat	No	decimal	VAT value
permitNote	No	string	Notes associated with permit in case they are necessary.

Possible error codes are 400 and 500 further described in the Developer portal link: https://external-gw-staging.easyparksystem.net/api/swagger-ui.html#/permit-resource/createPermitUsingPOST

Delete a permit application

Developer portal link:

https://external-gw-staging.easyparksystem.net/api/swagger-ui.html#/permit-resource/deleteApplicationsUsingPOST

Endpoints:

Staging	https://external-gw-staging.easyparksystem.net/api/permit/batch-delete
Production	https://external-gw.easyparksystem.net/api/permit/batch-delete

Method: `POST`

Header:

```
Content-Type: application/json
X-Authorization: Bearer idToken
```

Request JSON object

Type	Description
Array of strings	Array of uniqueId strings to be deleted. uniqueId as defined in the response of the Create method

Response JSON object
If successful, the response will just indicate `true`. Possible error codes are 400 and 500 further described in the Developer portal link:
https://external-gw-staging.easyparksystem.net/api/swagger-ui.html#/permit-resource/deleteApplicationsUsingPOST

Get a permit application using uniqueId

Developer portal link:

https://external-gw-staging.easyparksystem.net/api/swagger-ui.html#/permit-resource/findByUniqueIdUsingGET

Endpoints:

Staging	https://external-gw-staging.easyparksystem.net/api/permit/{uniqueId}
Production	https://external-gw.easyparksystem.net/api/permit/{uniqueId}

Method: `GET`

Header:

```
Content-Type: application/json
X-Authorization: Bearer idToken
```

Path parameters:

Parameter name	Required?	Type	Description
uniqueId	Yes	string	Unique identifier in the permit HUB, given by the Create method response

Response JSON object
If the uniqueId is not found, the response will be empty. If a permit is found it will return the following:

Parameter name	Type	Description

uniqueId	string	Unique identifier to be used in further operations of the Permit HUB
permitName	string	Permit name to be shown in user interaction.
validFrom	string	Datetime that indicates when the user is able to start utilizing the permit. Format is yyyy-MM-ddTHH:mm:ss.SSSZ (ISO 8601)
validTo	string	Datetime that indicates when the permit expires. Format is yyyy-MM-ddTHH:mm:ss.SSSZ (ISO 8601)
carLicenseNumber	string	License number without spaces or special characters, only one license plate per permit application. Either carLicenseNumber or phoneNumber is required to match the permit application to a vehicle or user.
phoneNumber	string	Format: +COUNTRYCODE and no leading zeros. Either carLicenseNumber or phoneNumber is required to match the permit application to a vehicle or user.
permitIdentifier	string	Two possible values are “USE_CARLICENSENUMBER” and “USE_PHONENUMBER” depending on how the application is to be found.
parkingareaNo	integer	HUB ID for parking area applicable for this permit application.
visibleForEnforcement	boolean	Value will be true for non-activated permits and false for activated permits.
discountRate	decimal	From 0 to 100, percentage value of discounted parking. Will be null for non activated permits.
maxDiscount	decimal	Monetary value of the maximum discount that can be applied with the discountRate. Only relevant for activated permits.
remainingParkingMinutes	integer	Will be set to a value >= 0 for activated limited permit, otherwise will be null.
priceExclVat	decimal	Price without VAT
priceNotSubjectToVat	decimal	Price not subject to VAT
priceTotal	decimal	Price total including VAT
priceVat	decimal	VAT value

permitNote	string	Notes associated with permit in case they are necessary.

Get a permit application using license plate

Developer portal link:

https://external-gw-staging.easyparksystem.net/api/swagger-ui.html#/permit-resource/findByLicensePlateUsingGET

Endpoints:

Staging	https://external-gw-staging.easyparksystem.net/api/permit/license-plate/{licenseplate}
Production	https://external-gw.easyparksystem.net/api/permit/license-plate/{licenseplate}

Method: `GET`

Header:

```
Content-Type: application/json
X-Authorization: Bearer idToken
```

Path parameters:

Parameter name	Required?	Type	Description
licensePlate	Yes	string	License number without spaces or special characters.

Response JSON object
If the licenseplate is not found, the response will be empty. If a permit is found it will return the following:

Parameter name	Type	Description
uniqueId	string	Unique identifier to be used in further operations of the Permit HUB
permitName	string	Permit name to be shown in user interaction.

validFrom	string	Datetime that indicates when the user is able to start utilizing the permit. Format is yyyy-MM-ddTHH:mm:ss.SSSZ (ISO 8601)
validTo	string	Datetime that indicates when the permit expires. Format is yyyy-MM-ddTHH:mm:ss.SSSZ (ISO 8601)
carLicenseNumber	string	License number without spaces or special characters, only one license plate per permit application. Either carLicenseNumber or phoneNumber is required to match the permit application to a vehicle or user.
phoneNumber	string	Format: +COUNTRYCODE and no leading zeros. Either carLicenseNumber or phoneNumber is required to match the permit application to a vehicle or user.
permitIdentifier	string	Two possible values are “USE_CARLICENSENUMBER” and “USE_PHONENUMBER” depending on how the application is to be found.
parkingareaNo	integer	HUB ID for parking area applicable for this permit application.
visibleForEnforcement	boolean	Value will be true for non-activated permits and false for activated permits.
discountRate	decimal	From 0 to 100, percentage value of discounted parking. Will be null for non activated permits.
maxDiscount	decimal	Monetary value of the maximum discount that can be applied with the discountRate. Only relevant for activated permits.
remainingParkingMinutes	integer	Will be set to a value >= 0 for activated limited permit, otherwise will be null.
priceExclVat	decimal	Price without VAT
priceNotSubjectToVat	decimal	Price not subject to VAT
priceTotal	decimal	Price total including VAT
priceVat	decimal	VAT value
permitNote	string	Notes associated with permit in case they are necessary.

Car switch a permit application

Developer portal link:

https://external-gw-staging.easyparksystem.net/api/swagger-ui.html#/permit-resource/carSwitchUsingPOST

Endpoints:

Staging	https://external-gw-staging.easyparksystem.net/api/permit/car-switch
Production	https://external-gw.easyparksystem.net/api/permit/car-switch

Method: `POST`

Header:

```
Content-Type: application/json
X-Authorization: Bearer idToken
```

Request JSON object

Parameter name	Required?	Type	Description
uniqueIdList	Yes	string array	Array of uniqueId strings to be changed
carLicenceNumber	Yes	string	New license number to be set to the permit applications. Format: without spaces or special characters, only one license plate per permit application.

Response JSON object
Like the Create method, the response will be an array of these permit applications, which will include all fields back as verification.

Parameter name	Required?	Type	Description
uniqueId	Yes	string	Unique identifier to be used in further operations of the Permit HUB
permitName	Yes	string	Permit name to be shown in user interaction.

validFrom	Yes	string	Datetime that indicates when the user is able to start utilizing the permit. Format is yyyy-MM-ddTHH:mm:ss.SSSZ (ISO 8601)
validTo	Yes	string	Datetime that indicates when the permit expires. Format is yyyy-MM-ddTHH:mm:ss.SSSZ (ISO 8601)
carLicenseNumber	Yes	string	New license number, changed by this method.
phoneNumber	No	string	Format: +COUNTRYCODE and no leading zeros.
permitIdentifier	Yes	string	Two possible values are “USE_CARLICENSENUMBER” and “USE_PHONENUMBER” depending on how the application is to be found.
parkingareaNo	Yes	integer	HUB ID for parking area applicable for this permit application.
visibleForEnforcement	Yes	boolean	Value will be true for non-activated permits and false for activated permits.
discountRate	Yes	decimal	From 0 to 100, percentage value of discounted parking. Will be null for non activated permits.
maxDiscount	No	decimal	Monetary value of the maximum discount that can be applied with the discountRate. Only relevant for activated permits.
remainingParkingMinutes	No*	integer	Will be set to a value >= 0 for activated limited permit, otherwise will be null.
priceExclVat	No	decimal	Price without VAT
priceNotSubjectToVat	No	decimal	Price not subject to VAT
priceTotal	No	decimal	Price total including VAT
priceVat	No	decimal	VAT value
permitNote	No	string	Notes associated with permit in case they are necessary.

Possible error codes are 400 and 500 further described in the Developer portal link: https://external-gw-staging.easyparksystem.net/api/swagger-ui.html#/permit-resource/carSwitchUsingPOST